

RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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ABSTRACT

Audio temporal grounding requires identifying which event occurrence satisfies a query. When opposite event orders coexist, a model can localize a real sequence yet answer the wrong relation. We introduce RelTwin, a fine-tuning framework that contrasts locally valid timestamp answers. Shared excerpts form opposite-order windows in one recording; inverse queries exchange their target assignments. Candidate supervision and query-exchange consistency train relation-specific selection, followed by set-level preferences over coverage, count, and boundaries. Across SpotSound-Bench, Clotho-Moment, and UnAV-100, RelTwin improves both SpotSound-A and TimeAudio, with mIoU gains of up to 4.23 and 14.48 points over the corresponding backbones. On controlled inverse-query development pairs, candidate supervision raises JointPairAcc by 31.25 percentage points over matched paired SFT. This advantage persists under altered silence timing, supporting improved query–window binding while inference retains direct timestamp generation.

Index Terms— Audio temporal grounding, timestamp answer contrasts, query–window binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. When sounds recur, recognizing their categories does not determine which occurrence answers the query. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions select different windows. Returning the same real window for both is a *query–window binding error*. Sequence supervision rewards each correct timestamp answer without explicitly contrasting it with another acoustically supported window under the same query.

RelTwin (Fig. 1) makes this competition explicit through *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows in one recording. Both have real acoustic support, but reversing the query exchanges their target assignments. The competing answer is the correct answer to the inverse query.

RelTwin trains the model’s timestamp-answer scores to favor the query-consistent window and couples the preferences under query reversal. The same competition extends to interval sets with different coverage, event counts, and boundaries. This answer-level formulation adapts SpotSound-A and TimeAudio while preserving direct generation at inference.

Evaluations on three benchmarks show localization gains for both backbones. Matched candidate-supervision experiments link answer competition to relation selection: identical answers to inverse queries fall from 37 to 1 out of 160 pairs, and the binding advantage persists when silence timing changes.

Accordingly, we evaluate the method at two levels: public benchmark scores test transfer to standard grounding, while inverse-query pairs test whether the model selects the requested occurrence. Timing intervention provides a controlled test of positional shortcuts.

2. RELATED WORK

Audio temporal grounding. Weakly supervised grounding learns from clip-level captions [3], while audio-language models enable text generation [4, 5, 6]. TimeAudio adds temporal markers and absolute time-aware encoding [7]; SpotSound interleaves timestamps with audio tokens and trains on absent-event queries [8]. RelTwin addresses query-dependent selection among *present* windows.

Compositional audio–language learning. CLAP aligns audio and text in a shared representation space [9]. Counterfactual caption learning separates alternative descriptions while preserving factual audio–text consistency [10]. T-CLAP contrasts temporal order [11]; CompA evaluates order and attribute binding and trains with composition-aware negatives [12]. CoSTALA combines spatio-temporal contrast, local alignment, and feature consistency [13]. RelTwin instead supervises complete timestamp answers: both candidates are grounded in the recording, and their target roles exchange with the query.

Hard negatives for temporal grounding. SHINE uses generated negative queries and saliency ranking for compositional video grounding [14]. RelTwin contrasts complete

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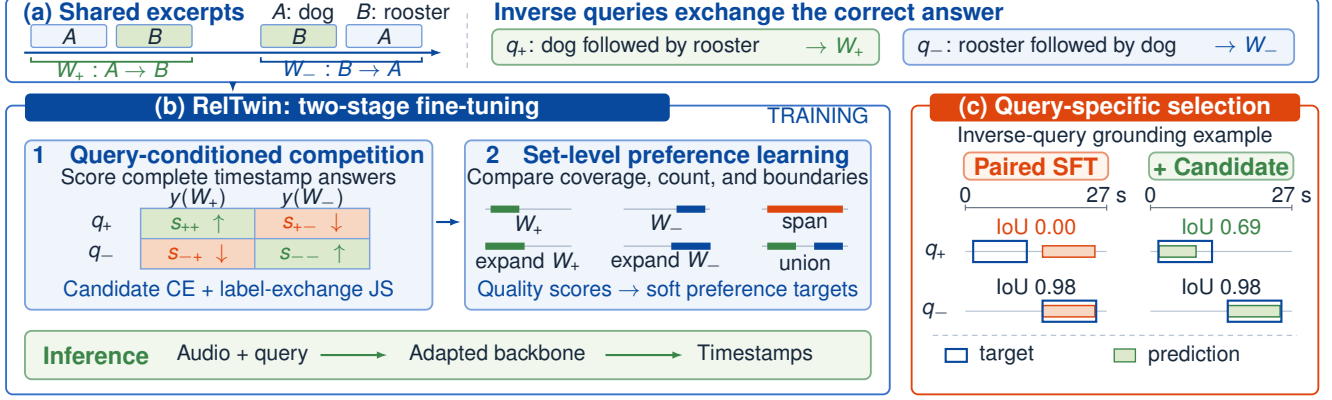


Fig. 1. RelTwin overview. (a) Shared excerpts hold acoustic content fixed while inverse queries exchange the target window. (b) Candidate supervision and query-exchange consistency train relation selection; set-level preference learning extends it to interval sets. Inference directly generates timestamps. Green/orange matrix cells indicate matching/competing answers. (c) A SpotSound-A inverse-query example compares paired SFT with candidate supervision. Blue outlines denote targets; filled bars denote predictions, with set-IoU above.

answer likelihoods in a fixed audio context, retaining the grounder’s direct generation.

3. RELTWIN

For recording x of duration T and query q , a generative grounder predicts $S = \{[a_k, b_k]\}_{k=1}^n$, with $0 \leq a_k < b_k \leq T$. RelTwin trains answer preferences in two stages: selecting the query-consistent occurrence, then refining interval-set quality.

3.1. Query-dependent twin construction

RelTwin arranges the same ESC-50 excerpts [15], A and B , as $A \rightarrow B$ and $B \rightarrow A$ in one recording. Targets $W_{\pm}$ span each sequence, including its gap; queries $q_{\pm}$ request the respective orders. Reusing excerpts holds acoustic content fixed: W_- is negative for q_+ but positive for q_- . We alternate which order appears first.

3.2. Answer competition and query-exchange consistency

We serialize S in the backbone’s native format: comma-separated spans for SpotSound-A and temporal tokens for TimeAudio. Answer tokens $y(S)$ exclude the prompt and receive the length-normalized score

$$s_{\theta}(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_{\theta}(y_t \mid x, q, y_{<t}). \quad (1)$$

The 2×2 matrix $M_{ij} = s_{\theta}(x, q_i, W_j)$ indexes queries by rows and answers by columns, $i, j \in \{+, -\}$ (Fig. 1b). Its diagonal entries are the correct query–answer pairs. Sequence supervision minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$. Candidate supervision additionally trains each diagonal score to exceed the

competing answer in its row. For $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$ with $\tau > 0$,

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

Query reversal exchanges the target assignment, so the two preference distributions should agree after swapping their labels. We enforce this correspondence with $\mathcal{L}_{\text{JS}} = \text{JS}(p_+, \Pi p_-)$, where Π swaps the two candidates. For an ordinary localization example (x_r, q_r, S_r) , rehearsal uses $\mathcal{R}_1 = -s_{\theta}(x_r, q_r, S_r)$. Stage 1 optimizes

$$\mathcal{L}_1 = \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{cand}} + \gamma \mathcal{L}_{\text{JS}}) + \beta \mathcal{R}_1. \quad (3)$$

3.3. Set-level preference learning

Stage 2 extends selection to coverage, count, and boundaries through six candidates: $W_{\pm}$, their expanded versions, their enclosing span, and their union. Expansions expose boundary excess; the span includes intervening background; the union adds the competing occurrence. Their relative quality becomes the answer-level training signal.

For candidate S_j and target W_i , a four-axis quality vector h_{ij} measures set-IoU, threshold margin, count agreement, and boundary accuracy. The threshold axis averages soft F1 at $\eta = 0.3/0.5$. Each interval’s best-match IoU u contributes $\max(0, u - \eta)/(1 - \eta)$; averaging over predictions and targets gives soft precision and recall. With n_S denoting interval count, count agreement is $1 - |n_{S_j} - n_{W_i}| / \max(n_{S_j}, n_{W_i}, 1)$. Boundary accuracy is $\exp(-e)$: e averages the summed endpoint error for each target’s best-IoU match, divided by $2 \max(b - a, 0.1)$ for target $[a, b]$.

Among K candidates, let d_{ij}^+ and d_{ij}^- count those dominated by and dominating S_j . Dominance means no worse on

Table 1. Temporal grounding across three benchmarks with RelTwin adaptation of TimeAudio and SpotSound-A (%; higher is better). Baseline results: SpotSound [8, Table 3]. Δ : gain over the corresponding backbone. Bold marks column maxima.

Method	SpotSound-Bench			Clotho-Moment			UnAV-100 subset		
	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU
<i>Non-LLM models</i>									
WTATG	53.0	34.5	38.4	12.1	6.3	9.1	53.0	37.0	38.4
AM-DETR	25.5	15.0	19.5	89.8	88.0	80.9	59.0	46.0	42.8
<i>Proprietary models</i>									
Gemini-2.5-Flash	34.8	27.8	25.7	45.7	33.7	36.9	48.0	37.0	35.6
Gemini-2.5-Pro	20.0	17.3	19.7	40.4	32.5	32.5	39.0	35.0	34.6
<i>Open-source models and RelTwin adaptation</i>									
Kimi-Audio	3.0	1.3	3.2	0.7	0.1	0.9	8.0	2.0	5.3
Qwen2-Audio	5.5	1.8	4.4	6.1	1.3	5.7	14.0	4.0	9.7
Audio Flamingo 3	8.8	2.8	7.6	32.9	21.8	22.6	35.0	24.0	25.0
TimeAudio	7.8	1.8	10.2	39.5	24.9	28.6	21.0	7.0	16.0
TimeAudio + RelTwin	24.25	13.25	18.36	45.63	30.92	34.08	44.00	25.00	30.48
Δ vs. TimeAudio	+16.45	+11.45	+8.16	+6.13	+6.02	+5.48	+23.00	+18.00	+14.48
SpotSound-A	76.5	59.5	57.9	93.4	91.0	85.6	86.0	74.0	69.8
SpotSound-A + RelTwin	78.00	62.00	59.43	93.61	91.01	86.68	89.00	79.00	74.03
Δ vs. SpotSound-A	+1.50	+2.50	+1.53	+0.21	+0.01	+1.08	+3.00	+5.00	+4.23

any axis and better on at least one. We define utility as

$$v_{ij} = \frac{1}{2} \left[\frac{1}{4} \sum_{\ell=1}^4 h_{ij\ell} + \frac{d_{ij}^+ - d_{ij}^-}{K-1} \right]. \quad (4)$$

Mean quality preserves graded differences; dominance rewards improvements without sacrificing another axis. Soft targets $t_i = \text{softmax}(v_i/\tau_q)$ guide the same answer scores, now normalized over all six candidates. Their listwise loss is $\mathcal{L}_{\text{set}} = -\frac{1}{2} \sum_{i,j} t_i(j) \log p_i(j)$. The six-way permutation $\Pi = (2, 1, 4, 3, 5, 6)$ swaps targets and expansions, fixing the span and union.

Replay candidates perturb boundaries, add/drop events, merge spans, split the longest interval, or cover the recording. The same quality rule defines their listwise loss $\mathcal{L}_{\text{set},r}$. A frozen base-model distribution p_0 anchors replay preferences p_r . Stage 2 optimizes

$$\begin{aligned} \mathcal{L}_2 &= \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{set}} + \gamma\mathcal{L}_{\text{JS}}) + \beta\mathcal{R}_2, \\ \mathcal{R}_2 &= \mathcal{R}_1 + \lambda\mathcal{L}_{\text{set},r} + \kappa \text{KL}(p_0||p_r). \end{aligned} \quad (5)$$

Both backbones share Eq. 1. Training updates LoRA parameters and, for TimeAudio, temporal/audio connectors. Inference remains direct timestamp generation from (x, q) , without twin construction or candidate scoring.

4. EXPERIMENTAL SETUP

Public benchmarks. We evaluate general localization on SpotSound-Bench (400 queries, 387 audios) [8], Clotho-Moment (6,649 queries), and UnAV-100 (100 queries, 77

audios) [2]. We report set-mIoU and R1 at IoU 0.3/0.5, clipping intervals to recording duration.

Twin data and relation evaluation. Forty ESC-50 classes yield 256 training recordings, 1,024 relation queries (512 pairs), and 512 category queries. Templates “followed by” and “and then” give two inverse-query pairs per recording. Relation development uses 10 held-out classes, 80 recordings, and 320 queries (160 pairs), with disjoint classes and original Freesound IDs at adaptation. It tests query-specific selection; timing intervention tests persistence under changed silence (Section 5.3).

Joint localization and relation selection. For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$ (zero for an empty union). Given predictions $\widehat{W}_i$, let $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$. Swap error occurs if $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. JointPairAcc requires PairAcc success without Swap error. Inspired by paired compositional evaluation [16], it distinguishes query-specific selection from coverage: returning the union of two disjoint, equal-length targets passes PairAcc (IoU 0.5) but fails JointPairAcc. We report pair-level percentages; same-answer counts require identical parsed interval lists.

RelTwin training. TimeAudio starts from its official checkpoint with 384/128 updates in Stages 1/2. Its 31.76M trainable parameters use learning rates 2×10^{-6} for LoRA [17] and 5×10^{-6} for temporal/audio connectors. Category queries target both occurrences of A or B ; category- and relation-query pairs enter the paired objective at 3:1. Replay mixes Clotho-Moment training [2], synthetic scale/cardinality, TEMPO training [18], and AudioGrounding training [1] at 4:3:2:1. We use $\lambda = \gamma = 1$, $\beta = 0.75$,

Table 2. Mechanism experiment: isolating RelTwin’s candidate supervision on SpotSound-A. (b) Matched data and updates; (c) mean $\pm$ sample SD. Joint: JointPairAcc; same-answer: inverse-query pairs with identical timestamp answers. Scores are percentages except counts.

Method	SpotSound	Relation development (160 inverse-query pairs)			
	mIoU $\uparrow$	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
<i>(a) Backbone reference: shared evaluation protocol</i>					
SpotSound-A	58.42	37.64	0.63	65.00	15/160
<i>(b) Matched training comparison</i>					
Paired SFT	58.42	74.58	53.13	40.00	37/160
Candidate supervision	59.43	88.55	84.38	7.50	1/160
Gain over paired SFT	+1.02	+13.97	+31.25	−32.50	−36 pairs
<i>(c) Candidate supervision: three training seeds</i>					
Candidate supervision	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46 $\pm$ 2.53	8.13 $\pm$ 0.63	(1, 0, 0)/160

$\kappa = 0.2$, $\tau = 1$, $\tau_q = 0.15$, and 15% interval expansion (minimum 0.05 s).

Matched mechanism experiment. Table 2 isolates $\mathcal{L}_{\text{cand}}$ on SpotSound-A after Stage 1: $\gamma = 0$, $\beta = 0.5$, and $\lambda = 0$ (paired SFT) or 1 (candidate supervision). Both share initialization, relation/rehearsal data, data order, and 256 AdamW [19] updates at 5×10^{-6} , with zero weight decay and gradient clipping 1. LoRA uses rank 8, $\alpha = 16$, dropout 0.1, and 20.19M parameters. Each update accumulates one pair and one rehearsal query. Candidate supervision adds four gradient-free scores and weighted gradient recomputation to SFT’s two positive-answer backpropagations.

Inference. Decoding is greedy, one beam, with at most 128 new tokens. The mechanism experiment uses 16 kHz mono audio and bfloat16.

Timing intervention. We recompose 80 development recordings with 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and 2 s tail, preserving event waveforms, order, and queries. Frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain waveform-identical.

Comparison and statistics. Table 2 evaluates the backbone and training variants under shared inference. TimeAudio uses seed 0. Mechanism point estimates use seed 0, with matched SFT; Table 2(c) separately reports seeds 0/1/2 as mean $\pm$ sample SD.

We use 20,000 audio-group bootstrap resamples for 95% CIs; relation tests resample 39 connected source-sharing groups (largest: 18 audios). The intervals quantify sampling variation for the selected checkpoints.

5. RESULTS AND ANALYSIS

5.1. Grounding performance across benchmarks

RelTwin improves both backbones on all three benchmarks (Table 1). The mIoU gains range from 1.08 to 4.23 points for SpotSound-A and 5.48 to 14.48 for TimeAudio. Adapted

SpotSound-A leads the listed methods. Gains on both backbones support adaptation across timestamp representations.

5.2. Timestamp contrasts recover query-specific selection

With matched data and updates, candidate supervision improves SpotSound mIoU by 1.02 points over paired SFT (95% audio-group CI: [0.23, 1.81]; Table 2).

JointPairAcc rises by 31.25 points (source-group CI [22.35, 40.50]). Identical-answer pairs fall from 37 to 1 out of 160; 27 of those 37 become jointly correct. Figure 1 illustrates this recovery. Across all pairs, transitions to joint correctness outnumber regressions by 51 to 1.

The relation mIoU gain is 13.97 points (CI [9.84, 17.51]). Three-seed mean JointPairAcc is 81.46, with same-answer counts of 1, 0, and 0. Both windows contain the queried categories, so recovering distinct, jointly correct answers demonstrates relation selection beyond category presence.

5.3. Binding gains persist after timing intervention

On the 40 recordings with altered silence timing, candidate supervision retains a 20-point JointPairAcc advantage (87.50 vs. 67.50; CI [2.94, 33.33]), compared with original scores of 85.00 and 42.50. The advantage persists when event waveforms, order, and queries are fixed but silence timing changes.

Together, public gains test transfer, paired tests isolate selection, and timing changes test position-independent relation learning; set-level learning extends this decision to coverage and boundaries.

6. CONCLUSION

RelTwin turns competing, acoustically supported answers into supervision for query-specific grounding. Inverse queries exchange which answer is correct. Gains across two backbones and three benchmarks, together with matched comparisons and timing intervention, support this answer-level learning approach while preserving direct generation.

7. ACKNOWLEDGMENTS

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Compliance with Ethical Standards. We use public ESC-50 [15], SpotSound [8], Clotho-Moment, UnAV [2], TEMPO, and AudioGrounding data, including programmatically constructed sequences. The study involved no new recordings, participant recruitment, human or animal experiments, or listening studies.

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